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PERFORMANCE SPECIFICATION

DECK COVERING MATERIALS, MONOLITHIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers monolithic deck covering materials for interior shipboard applications.

1.2 Classification. Monolithic deck covering materials are of the following types, classes, grades, and compositions as specified (see 6.2).

1.2.1 Types.

a. Type I – Coating, high durability, general

b. Type II – Coating, high durability, corrosion resistant

c. Type III – Coating, aqueous film forming foam (AFFF) resistant

d. Type IV – Chemical-resistant floorings (resin-based seamless flooring [e.g., cosmetic polymeric, terrazzo, and color flake]), underlayment required

e. Type V – Chemical-resistant floorings (resin-based seamless flooring [e.g., cosmetic polymeric, terrazzo, and color flake]), no underlayment required

f. Type VI – Latex concrete

1.2.2 Classes.

a. Class 1 – For general shipboard use

b. Class 2 – For use on submarines

1.2.3 Grades.

a. Grade A – All system components of the qualifying system have a volatile organic compound (VOC) not greater than 250 grams per liter (2.08 pounds per gallon).

b. Grade B – All system components of the qualifying system have a VOC not greater than 340 grams per liter (2.84 pounds per gallon).

1.2.4 Compositions.

a. Composition E – Epoxy

b. Composition P – Polyurethane or polyurea

c. Composition O – Other

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-3135	-	Deck Covering Underlay Materials
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-23236	-	Coating Systems for Ship Structures
MIL-PRF-24385	-	Fire Extinguishing Agent, Aqueous Film-Forming Foam (AFFF) Liquid Concentrate, for Fresh and Sea Water
MIL-DTL-24441	-	Paint, Epoxy-Polyamide, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1623	-	Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 60, Appendix A-7, Method 24	-	Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings
40 CFR 63, Subpart II	-	National Emission Standards for Shipbuilding and Ship Repair (Surface Coating)
40 CFR 63, Appendix A, Method 311	-	Analysis of Hazardous Air Pollutant Compounds in Paints and Coatings by Direct Injection Into a Gas Chromatograph
40 CFR 261.24	-	Toxicity Characteristic

(Copies of these documents are available online at www.ecfr.gov.)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual (SACM)

(Copies of the chapter titled "Material Control Program" are available by email request to CommandStandards@navy.mil.)

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A326.3 - Dynamic Coefficient of Friction (DCOF) of Hard Surface Flooring Materials

(Copies of this document are available online at https://webstore.ansi.org/.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-SP 10/NACE NO. 2 - Near-White Metal Blast Cleaning

(Copies of this document are available online at www.ampp.org.)

ASTM INTERNATIONAL

ASTM A568/A568M	-	Standard Specification for Steel, Sheet, Carbon, Structural, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for
ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM C413	-	Standard Test Method for Absorption of Chemical-Resistance Mortars, Grouts, Monolithic Surfacings, and Polymer Concretes
ASTM C722	-	Standard Specification for Chemical-Resistant Monolithic Floor Surfacings
ASTM D257	-	Standard Test Methods for DC Resistance or Conductance of Insulating Materials
ASTM D523	-	Standard Test Method for Specular Gloss
ASTM D610	-	Standard Practice for Evaluating Degree of Rusting on Painted Steel Surfaces
ASTM D823	-	Standard Practices for Producing Films of Uniform Thickness of Paint, Varnish, and Related Products on Test Panels
ASTM D1141	-	Standard Practice for the Preparation of Substitute Ocean Water
ASTM D1193	-	Standard Specification for Reagent Water
ASTM D1308	-	Standard Test Method for Effect of Household Chemicals on Clear and Pigmented Organic Finishes

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ASTM D1640/D1640M	-	Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings
ASTM D1849	-	Standard Test Method for Package Stability of Paint
ASTM D2047	-	Standard Test Method for Static Coefficient of Friction of Polish-Coated Flooring Surfaces as Measured by the James Machine
ASTM D2794	-	Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM D3278	-	Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus
ASTM D3363	-	Standard Test Method for Film Hardness by Pencil Test
ASTM D4060	-	Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser
ASTM D4417	-	Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel
ASTM D6450	-	Standard Test Method for Flash Point by Continuously Closed Cup (CCCFP) Tester
ASTM E1252	-	Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis
ASTM F718	-	Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AMS-STD-595	-	Colors Used in Government Procurement
SAE J300	-	Engine Oil Viscosity Classification

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Deck covering materials furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material. Deck covering materials supplied under this specification shall comply with all Navy occupational health and safety regulations and meet the following characteristics.

3.2.1 Fire performance. When tested in accordance with 4.6, the deck covering system shall conform to the requirements in MIL-STD-1623.

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3.2.2 Off-gassing (class 2 only). The class 2 deck covering system materials shall be evaluated for off-gassing in accordance with the requirements of 4.7. Based on the circumstances of use and the chemical nature of the class 2 deck covering system materials, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the class 2 deck covering system materials shall be assigned to either the "Permitted" or "Limited" category (see 4.7 and 6.6).

3.2.3 Toxicity and prohibited materials.

3.2.3.1 Toxicity. When evaluated in accordance with 4.8.1, the deck covering system shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.8.1 and 6.7).

3.2.3.2 Prohibited materials. When evaluated in accordance with 4.8.2, the deck covering system shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.2.4 Metals content. When tested as specified in 4.9, the content of each soluble metal and the total content in each system component of the deck covering system shall not exceed the values listed in table I. Total metal content values may be submitted to satisfy soluble metal content requirements when the total metals value is converted to milligrams per liter and determined to be lower than the maximum soluble metal content limits appearing in table I.

TABLE I. Metals content.

Metal and its compound in each deck covering dry film	Soluble metal, maximum (milligrams per liter)	Total metal content, maximum (weight percent)
Antimony	15	0.015
Arsenic	5	0.005
Barium (excluding barite)	100	0.10
Beryllium	0.75	0.0002
Cadmium	1	0.0005
Chromium (VI)	1	0.001
Chromium or chromium (III)	560	0.56
Cobalt	50	0.005
Copper	25	0.01
Fluoride salts	180	0.18
Lead	5	0.005
Mercury	0.2	0.0002
Molybdenum	350	0.35
Nickel	20	0.02
Selenium	1	0.002
Silver	5	0.001
Tantalum	100	0.100
Thallium	7	0.007
Tungsten	100	0.100
Vanadium	24	0.01
Zinc	250	0.25

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3.2.5 VOC content.

3.2.5.1 Grade A. When tested as specified in 4.10, the VOC content of each system component of the deck covering system shall not exceed 250 grams per liter (2.08 pounds per gallon).

3.2.5.2 Grade B. When tested as specified in 4.10, the VOC content of each system component of the deck covering system shall not exceed 340 grams per liter (2.84 pounds per gallon).

3.2.6 Hazardous air pollutant (HAP) content. When evaluated as specified in 4.11, the total HAP content, as defined by the 40 CFR 63, Subpart II (Environmental Protection Agency [EPA] National Emission Standards for Hazardous Air Pollution [NESHAP]) requirements for shipbuilding and ship repair, of each system component, shall not exceed the VOC limit prescribed for the specific grades (see 3.2.5).

3.3 Flash point. When tested as specified in 4.12, all liquid components of the coating system shall flash at a temperature not lower than 38 °C (100 °F).

3.4 Identification (ID) characteristics. Values for ID characteristics shall be those established for the product, including all variable components (e.g., all available colors and manufacturers of color chips), at the time it is submitted for qualification testing. The purpose of these values is to serve as a basis for determining that the coating or coating component being offered is essentially the same as that which was approved under qualification testing. Unless otherwise specified herein, ID characteristics for all types, classes, grades, and compositions shall be conducted at 16 to 27 °C (60 to 80 °F) and below 85 percent relative humidity. ASTM methods, other consensus standards, or a copy of unique test methodology or any other necessary information on methodology used to determine reported ID characteristics shall be provided as specified (see 6.2).

a. SAE AMS-STD-595 or other identified color and pattern (excluding type VI)
b. Unit of issue
c. Maximum application thickness
d. Coating weight at maximum application thickness per unit area
e. Density for all liquid coating components
f. System parts (e.g., base/color coat, color chips, aggregate, sealer coat)
g. Chemical nature of each system component and part (e.g., epoxy, polyurethane, polyurea)
h. Application method(s)
i. Volumetric mix ratios of each system component
j. Aggregate or broadcast composition
k. Package stability of paint
l. Volume solids (excluding type VI)
m. Pot life
n. Viscosity of all liquid components and mixed deck coating
o. Manufacturer of each system part

3.5 Manufacturer's instructions for use (ASTM F718 product data sheet). As specified (see 6.2), manufacturer's instructions for use shall be provided in the ASTM F718 product data sheet for each system component and part. These instructions shall address use of non-coating parts, such as color chips and aggregate.

3.6 Requirements for types I, II, and III.

3.6.1 Package stability of paint. When tested as specified in 4.14.1, the viscosity of the mixed deck covering system shall be within the viscosity range identified in 3.4n, and the deck coating shall be rated not less than level 8 for any of the following parameters addressed in ASTM D1849:

a. Skinning, pressure, corrosion of the container, and odor of spoilage
b. Rigidity of the lower layer

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c. Grains, lumps, or streaks in the brushed film

3.6.2 Gloss. When tested as specified in 4.14.2, the gloss of the coating shall be greater than 45.

3.6.3 Abrasion resistance. When tested as specified in 4.14.3, the average weight loss shall not exceed 0.16 gram (0.006 ounce).

3.6.4 Impact resistance. When tested as specified in 4.14.4, the coating shall show no visible signs (without magnification) of chipping, cracking, or detachment from the test panel. The impact required to cause coating failure at the point of impact shall be not less than 3.16 Newton-meters (28 inch-pounds).

3.6.5 Slip resistance. When tested as specified in 4.14.5, the static coefficient of friction (SCOF) shall be not less than 0.70 for dry surfaces and 0.60 for wet surfaces.

3.6.6 Chemical resistance. When tested as specified in 4.14.6, the exposed coating system shall be visually indistinguishable from the panel before exposure.

3.6.7 AFFF resistance (type III only). When tested as specified in 4.14.7, the coating shall be visually indistinguishable from the panel before immersion and shall not decrease more than one unit of hardness.

3.6.8 Salt spray (fog) resistance (type II only). When tested as specified in 4.14.8, the coating on all test panels shall exhibit a rust grade not less than 8, in accordance with ASTM D610, where the rust grade does not exceed 0.1 percent rusted surface area.

3.7 Requirements for types IV and V.

3.7.1 Chemical-resistant monolithic floorings. When tested as specified in 4.15.1, the deck covering shall meet all the requirements of ASTM C722 as specified by application method classification, except as specified in 3.7.2.

3.7.2 Slip resistance. When tested as specified in 4.15.1, the deck covering SCOF shall be not less than 0.70 for dry surfaces and 0.60 for wet surfaces.

3.7.3 Chemical resistance. When tested as specified in 4.15.2, the exposed deck covering system shall be visually indistinguishable and shall retain at least 90 percent of the gloss of the panel before exposure.

3.7.4 Corrosion resistance. When tested as specified in 4.15.3, the deck covering shall not become detached, and the surface of the steel beneath the deck covering material shall show no signs of corrosion, wicking, or other water penetration.

3.7.5 Impact resistance. When tested as specified in 4.15.4, the deck covering shall show no visible signs of chipping, cracking, or detachment from the test panel. The permanent indentation of the deck covering shall be not greater than 1.6 millimeters (1/16 inch).

3.7.6 Indentation resistance. When tested as specified in 4.15.5, the initial indentation shall not exceed 5 percent of the deck covering thickness.

3.7.7 Vertical sag (type V only). When tested as specified in 4.15.6, the decking material at the bottom of the panel shall not exceed 150 percent of the thickness of the material at the top of the panel.

3.7.8 Abrasion resistance. When tested as specified in 4.15.7, the average weight loss shall not exceed 0.16 gram (0.006 ounce).

3.8 Requirements for type VI.

3.8.1 Indentation resistance. When tested as specified in 4.16.1, the initial indentation shall not exceed 1.5 percent of the deck covering thickness and shall show no signs of cracking or becoming detached from the test panel as a result of the indentation.

3.8.2 Impact resistance. When tested as specified in 4.16.2, the deck covering shall show no visible signs (without magnification) of chipping, cracking, or detachment from the test panel.

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3.8.3 Moisture absorption. When tested as specified in 4.16.3, the deck covering shall not gain more than 2 percent of the original weight of the specimen.

3.8.4 Bond strength. When tested as specified in 4.16.4, the average bond strength shall be not less than 14.6 kilograms per square centimeter (200 pounds per square inch).

3.8.5 Live load resistance. When tested as specified in 4.16.5, the deck covering shall show no signs of cracking or becoming detached from the steel base plates and shall show an indentation not greater than 1.3 millimeters (0.05 inch).

3.8.6 Slip resistance. When tested as specified in 4.16.6, the SCOF shall be not less than 0.70 for dry surfaces and 0.60 for wet surfaces.

3.8.7 Sparking resistance. When tested as specified in 4.16.7, the deck covering shall show no evidence of sparking.

3.8.8 Electrical resistance. When tested as specified in 4.16.8, the average surface resistivity shall not exceed 3 megohms.

3.8.9 Abrasion resistance. When tested as specified in 4.16.9, the material loss shall not exceed 0.16 gram (0.006 ounce).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of all the tests listed in table II.

4.2.1 Change approval. Changes in construction, formulation, materials, design, or manufacturing process of qualified products may require requalification. Unless otherwise specified by the qualifying activity, changes require written approval by NAVSEA.

4.3 Conformance inspection. When specified (see 6.2), conformance inspection shall consist of the conformance tests specified in table II. At a minimum, conformance inspections shall be performed on the first lot and every 19,000 liters (5,000 gallons) thereafter. For the purpose of conformance inspection, a lot shall consist of a liquid monolithic deck covering component of the same type, class, grade, and composition from a single, uniform batch or a uniform blend of batches offered for delivery at one time. For each liquid component available in multiple colors, conformance testing of the first lot and every 19,000 liters (5,000 gallons) thereafter shall be determined by the cumulative, total volumes of all colors for that part produced by the manufacturer. Conformance testing is only required on one color of a given part.

4.3.1 Lot. A lot shall consist of an individual system component of the same type, class, grade, and composition from a single uniform batch or a uniform blend of batches offered for delivery at one time.

4.4 Inspection conditions. Unless otherwise specified herein, all physical tests (including preparation, application, and testing) on deck covering system materials shall be made at 16 to 27 °C (60 to 80 °F) and below 85 percent relative humidity.

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TABLE II. Test procedures.

Deck covering type	Characteristic	Requirement paragraph	Test paragraph	Applicable test standard	Conformance test
All	Fire performance	3.2.1	4.6	MIL-STD-1623	---
	Off-gassing(class 2 only)	3.2.2	4.7	---	---
	Toxicity and prohibited materials	3.2.3	4.8	---	---
	Metals content	3.2.4	4.9	---	---
	VOC content	3.2.5	4.10	---	X
	HAP content	3.2.6	4.11	---	---
	Flash point (liquid components only) (excluding type VI)	3.3	4.12	ASTM D3278/ ASTM D6450	X
	ID characteristics	3.4	4.5	---	---
	Volume solids (excluding type VI)	3.4	4.5	---	X
Types I, II, and III	Package stability of paint	3.6.1	4.14.1	ASTM D1849	---
	Gloss	3.6.2	4.14.2	ASTM D523	X
	Abrasion resistance	3.6.3	4.14.3	ASTM D4060	---
	Impact resistance	3.6.4	4.14.4	ASTM D2794	X
	Slip resistance	3.6.5	4.14.5	ASTM D2047	X
	Chemical resistance	3.6.6	4.14.6	ASTM D1308	---
	AFFF resistance (type III only)	3.6.7	4.14.7	ASTM D1308	---
	Salt spray (fog) resistance (type II only)	3.6.8	4.14.8	ASTM B117/ ASTM D610	---
Types IV and V	Chemical-resistant monolithic floor surfacing	3.7.1	4.15.1	ASTM C722	---
	Slip resistance	3.7.2	4.15.1	ASTM C722/ ASTM D2047	---
	Chemical resistance	3.7.3	4.15.2	ASTM C722/ ASTM D523	---
	Corrosion resistance	3.7.4	4.15.3	ASTM C722	---
	Impact resistance	3.7.5	4.15.4	--	X
	Indentation resistance	3.7.6	4.15.5	---	---
	Vertical sag (type V only)	3.7.7	4.15.6	---	---
	Abrasion resistance	3.7.8	4.15.7	ASTM D4060	---

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TABLE II. Test procedures – Continued.

Deck covering type	Characteristic	Requirement paragraph	Test paragraph	Applicable test standard	Conformance test
Type VI	Indentation resistance	3.8.1	4.16.1	---	---
	Impact resistance	3.8.2	4.16.2	---	X
	Moisture absorption	3.8.3	4.16.3	ASTM C413	---
	Bond strength	3.8.4	4.16.4	---	---
	Live load resistance	3.8.5	4.16.5	---	---
	Slip resistance	3.8.6	4.16.6	ASTM D2047	X
	Sparking resistance	3.8.7	4.16.7	---	---
	Electrical resistance	3.8.8	4.16.8	ASTM D257	---
	Abrasion resistance	3.8.9	4.16.9	---	---

4.5 General inspection. The deck covering system shall be visually inspected, and the ID characteristics shall be reviewed to ensure conformance to the requirements specified in 3.4.

4.6 Fire performance. The as-applied deck covering system shall be tested in accordance with MIL-STD-1623 on 0.64-centimeter- (¼-inch-) thick steel or aluminum panels, or fiber-cement board having a density of 1,442±160 kilograms per cubic meter (90±10 pounds per cubic foot). Results of fire performance testing shall conform to the requirements of 3.2.1 and 4.6.1.

4.6.1 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to NAVSEA. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.5).

4.7 Off-gassing (class 2 only). The class 2 deck covering system materials shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled "Material Control Program" (see 3.2.2 and 6.6). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA-approved test facility (see 3.2.2). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.2.2).

4.8 Toxicity and prohibited materials.

4.8.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.3.1 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the deck covering system using data provided by the manufacturer/distributor to the NMCPHC (see 3.2.3.1 and 6.7).

4.8.2 Prohibited materials. Prohibited materials shall be verified for conformance to 3.2.3.2 as required by the qualifying activity.

4.9 Metals content. Soluble and total metal content, except tantalum and tungsten, shall be determined on pulverized cured film of the deck covering system in accordance with 40 CFR 261.24(a), Toxicity Characteristic Leaching Procedure (TCLP). The test results for each metal shall be as specified in 3.2.4. Tantalum and tungsten soluble metal content and total metal content shall be analyzed as specified in 4.9.1. Calculation of individual hazardous metal contents can be based on either the manufacturer's testing of batches or the supplier's data for raw materials used in the product. When specified (see 6.2), a formulation value shall be provided that shall not be exceeded when the deck covering is tested in accordance with this paragraph.

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4.9.1 Tantalum and tungsten content. The tantalum and tungsten content of the cured deck covering shall be determined using any appropriate spectroscopy test method. The tests shall be conducted in accordance with the instrument manufacturer's directions. Data supporting the test method choice and analytical accuracy shall be established. The test results for tantalum or tungsten shall be in conformance with the requirements of 3.2.4.

4.10 VOC content. The VOC content of each system component shall be measured in accordance with 40 CFR 60, Appendix A-7, Method 24. Results shall conform to the requirements of 3.2.5.

4.11 HAP content. The HAP content of each system component shall be measured in accordance with 40 CFR 63, Appendix A, Method 311 (EPA Test Method 311). Solvent fractions shall be identified in accordance with ASTM E1252, with the results recorded as percent weight of the total covering. Alternate methods of analysis shall be approved by the qualifying activity. When specified by the qualifying activity, formulation data may be used by manufacturers in lieu of testing to demonstrate compliance with 3.2.6. The formulation data shall have a consistent and quantitatively known relationship to the testing required. Calculation of individual HAP contents can be based on either manufacturer evaluation of batches or supplier data for raw materials used in the product. Results shall conform to the requirements of 3.2.6.

4.12 Flash point. All deck covering materials shall be tested in accordance with ASTM D3278, ASTM D6450, or through a method approved by the qualifying activity. Results shall conform to the requirements of 3.3.

4.13 Test panel preparation.

4.13.1 Method 1 (types I, II, and III). The test panel shall be ground glass, measuring 7.5 centimeters by 15 centimeters (3 inches by 6 inches), a Leneta© Form 7B Sag and Leveling Test Chart, or an equivalent test chart with a measurement surface at least 7.5 centimeters by 15 centimeters (3 inches by 6 inches). The topcoat of the coating system, color, or sealer coat shall be conditioned and mixed in accordance with the ASTM F718 product data sheet. A film of mixed coating shall be drawn in accordance with ASTM D823, Practice E, to produce a wet film thickness of 75±25 micrometers (3±1 mils).

4.13.2 Method 2 (types I, II, and III). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 10.2 centimeters by 10.2 centimeters by 3.2 millimeters (4 inches by 4 inches by ⅛ inch) with a 6.4-millimeter (¼-inch) hole in the center shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. The deck coating shall be conditioned and mixed in accordance with the ASTM F718 product data sheet. Two coats of the color coat shall be applied to all exposed surfaces in accordance with the ASTM F718 product data sheet to produce a total dry film thickness (DFT) of 508±102 micrometers (20±4 mils). The final coat shall be cured at 23±2.5 °C (73±5 °F) for 7 days, or until full cure-to-service time in accordance with the ASTM F718 product data sheet is reached, whichever is shorter. If a sealer coat is part of the qualifying system, it shall be applied in accordance with the ASTM F718 product data sheet as a third coat. If color chips or broadcast aggregate are being qualified as part of the system, do not apply to these panels.

4.13.3 Method 3 (types I, II, and III). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 15.2 centimeters by 30.5 centimeters by 3.2 millimeters (6 inches by 12 inches by ⅛ inch) (or 23 centimeters by 28 centimeters by 3.2 millimeters [9 inches by 11 inches by ⅛ inch] for ASTM D2047 or 30.5 centimeters by 30.5 centimeters by 3.2 millimeters [12 inches by 12 inches by ⅛ inch] for ASTM C1038) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. The deck coating shall be conditioned and mixed in accordance with the ASTM F718 product data sheet. Two coats of the color coat shall be applied to all exposed surfaces in accordance with the ASTM F718 product data sheet to produce a total DFT of 508±102 micrometers (20±4 mils). The final coat shall be cured at 23±2.5 °C (73±5 °F) for 7 days, or until full cure-to-service time in accordance with the ASTM F718 product data sheet is reached, whichever is shorter. If color chips, aggregate, or sealer coat are being qualified as part of the system, they shall be applied in accordance with the ASTM F718 product data sheet over the two color coats.

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4.13.4 Method 4 (types I, II, and III). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 15.2 centimeters by 30.5 centimeters by 6.4 millimeters (6 inches by 12 inches by ¼ inch) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. The deck coating shall be conditioned and mixed in accordance with the ASTM F718 product data sheet. Two coats of the color coat shall be applied to all exposed surfaces in accordance with the ASTM F718 product data sheet to produce a total DFT of 508±102 micrometers (20±4 mils). The final coat shall be cured at 23±2.5 °C (73±5 °F) for 7 days, or until full cure-to-service time in accordance with the ASTM F718 product data sheet is reached, whichever is shorter. If color chips, aggregate, or sealer coat are being qualified as part of the system, they shall be applied in accordance with the ASTM F718 product data sheet over the two color coats.

4.13.5 Method 5 (types IV and V). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 15.2 centimeters by 30.5 centimeters by 3.2 millimeters (6 inches by 12 inches by ⅛ inch) (or 23 centimeters by 28 centimeters by 3.2 millimeters [9 inches by 11 inches by ⅛ inch] for ASTM D2047) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch), as measured per ASTM D4417, method B or C. The complete qualifying system shall be conditioned and applied in accordance with the ASTM F718 product data sheet to the maximum application thickness identified in 3.4.c.

4.13.6 Method 6 (type IV). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 15.2 centimeters by 30.5 centimeters by 3.2 millimeters (6 inches by 12 inches by ⅛ inch) (or 23 centimeters by 28 centimeters by 3.2 millimeters [9 inches by 11 inches by ⅛ inch] for ASTM D2047) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch), as measured per ASTM D4417, method B or C. A deck primer (qualified to MIL-DTL-24441, type IV or MIL-PRF-23236, type V or VI) and underlayment system (qualified to MIL-PRF-3135, type I or III, class 2) or a direct-to-metal underlayment system (qualified to MIL-PRF-3135, type IV, class 2) shall be applied to a thickness of 6.4 millimeters (¼ inch) in accordance with the ASTM F718 product data sheet. Once the underlayment has reached the manufacturer-specified cure time, the complete qualifying system shall be conditioned and applied in accordance with the ASTM F718 product data sheet to the maximum application thickness identified in 3.4.c.

4.13.7 Method 7 (type V). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 15.2 centimeters by 30.5 centimeters by 6.4 millimeters (6 inches by 12 inches by ¼ inch) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. The complete qualifying system shall be conditioned and applied in accordance with the ASTM F718 product data sheet to the maximum application thickness identified in 3.4.c.

4.13.8 Method 8 (type IV). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 15.2 centimeters by 30.5 centimeters by 6.4 millimeters (6 inches by 12 inches by ¼ inch) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. A deck primer (qualified to MIL-DTL-24441, type IV or MIL-PRF-23236, type V or VI) and underlayment system (qualified to MIL-PRF-3135, type I or III, class 2) or a direct-to-metal underlayment system (qualified to MIL-PRF-3135, type IV, class 2) shall be applied to a thickness of 6.4 millimeters (¼ inch) in accordance with the ASTM F718 product data sheet. Once the underlayment has reached the manufacturer-specified cure time, the complete qualifying system shall be conditioned and applied in accordance with the ASTM F718 product data sheet to the maximum application thickness identified in 3.4.c.

4.13.9 Method 9 (type V). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 10.2 centimeters by 10.2 centimeters by 3.2 millimeters (4 inches by 4 inches by ⅛ inch) with a 6.4-millimeter (¼-inch) hole in the center shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation. The complete qualifying system shall be conditioned and applied in accordance with the ASTM F718 product data sheet. If a sealer coat is part of the qualifying system, it shall be applied in accordance with the ASTM F718 product data sheet as a third coat. If color chips or broadcast aggregate are being qualified as part of the system, do not apply to these panels.

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4.13.10 Method 10 (type IV). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 10.2 centimeters by 10.2 centimeters by 3.2 millimeters (4 inches by 4 inches by ⅛ inch) with a 6.4-millimeter (¼-inch) hole in the center shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. A deck primer (qualified to MIL-DTL-24441, type IV or MIL-PRF-23236, type V or VI) and underlayment system (qualified to MIL-PRF-3135, type I or III, class 2) or a direct-to-metal underlayment system (qualified to MIL-PRF-3135, type IV, class 2) shall be applied to a thickness of 3.2 millimeters (⅛ inch) in accordance with the ASTM F718 product data sheet. Once the underlayment has reached the manufacturer-specified cure time, the qualifying system shall be conditioned and applied in accordance with the ASTM F718 product data sheet. If a sealer coat is part of the qualifying system, it shall be applied in accordance with the ASTM F718 product data sheet as a third coat. If color chips or broadcast aggregate are being qualified as part of the system, do not apply to these panels.

4.13.11 Method 11 (type VI). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 30.5 centimeters by 30.5 centimeters by 6.4 millimeters (12 inches by 12 inches by ¼ inch) (or 23 centimeters by 28 centimeters by 6.4 millimeters [9 inches by 11 inches by ¼ inch] for ASTM D2047) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. If required by the ASTM F718 product data sheet, the panel shall be primed using a coating (qualified to MIL-DTL-24441, type IV or MIL-PRF-23236, type V or VI). The complete deck covering system shall be conditioned and applied in accordance with the ASTM F718 product data sheet to the maximum application thickness identified in 3.4.c. If no maximum application thickness is identified, the deck covering shall be applied to meet a DFT of 5.1 centimeters (2 inches).

4.13.12 Method 12 (type VI). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 5.1 centimeters by 15 centimeters by 3.2 millimeters (2 inches by 6 inches by ⅛ inch) shall be prepared by applying the deck covering in an area of 5.1 centimeters by 5.1 centimeters (2 inches by 2 inches) at a thickness of 1.3 centimeters (½ inch), or in accordance with the ASTM F718 product data sheet to the maximum system thickness identified in 3.4.c, such that 2.54 centimeters (1 inch) of the plate is exposed at one end of the panel and 7.6 centimeters (3 inches) is exposed at the opposite end of the panel. If required by the ASTM F718 product data sheet, the panel shall be primed using a single primer coat (qualified to MIL-DTL-24441, type IV or MIL-PRF-23236, type V or VI). The complete deck covering system shall be conditioned, mixed, and allowed to cure in accordance with the ASTM F718 product data sheet.

4.13.13 Method 13 (type VI). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 23 centimeters by 33 centimeters by 6.4 millimeters (9 inches by 13 inches by ¼ inch) shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. If required by the ASTM F718 product data sheet, the panel shall be primed using a coating (qualified to MIL-DTL-24441, type IV or MIL-PRF-23236, type V or VI). The complete deck covering system shall be conditioned and applied in accordance with the ASTM F718 product data sheet to the dimensions of 20 centimeters by 25.4 centimeters by 5.1 centimeters (8 inches by 10 inches by 2 inches).

4.13.14 Method 14 (type VI). Mild steel panels conforming to ASTM A568/A568M, cold rolled, measuring approximately 7.6 centimeters by 5.0 centimeters by 3.2 millimeters (3 inches by 2 inches by ⅛ inch) with a 6.4-millimeter (¼-inch) hole in the center shall be prepared in accordance with SSPC-SP 10/NACE NO. 2 surface preparation to provide an average profile pattern of 0.051 to 0.076 millimeter (0.002 to 0.003 inch) as measured per ASTM D4417, method B or C. If required by the ASTM F718 product data sheet, the panel shall be primed using a coating (qualified to MIL-DTL-24441, type IV or MIL-PRF-23236, type V or VI). The complete deck covering system shall be conditioned and applied in accordance with the ASTM F718 product data sheet to the maximum application thickness identified in 3.4.c. If no maximum application thickness is identified, the deck covering shall be applied to meet a DFT of 5.1 centimeters (2 inches).

4.14 Test procedures for types I, II, and III.

4.14.1 Package stability of paint. The deck coating shall be tested in accordance with ASTM D1849 and stored at 60±2.5 °C (140±5 °F) for a period of 14 days to determine conformance to 3.6.1.

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4.14.2 Gloss. One test panel shall be prepared in accordance with 4.13.1. The test panel shall be dry-hard in accordance with ASTM D1640/D1640M, method A, at 23±2.5 °C (73±5 °F) prior to testing. The 60-degree specular gloss of the cured coating shall be determined in accordance with ASTM D523. The results shall conform to 3.6.2.

4.14.3 Abrasion resistance. Three test panels shall be prepared in accordance with 4.13.2. Abrasion resistance shall be tested in accordance with ASTM D4060 using a CS-17 wheel and a 1-kilogram (2.2-pound) load for 1,000 cycles. The results shall conform to 3.6.3.

4.14.4 Impact resistance. Three test panels shall be prepared in accordance with 4.13.4. Impact resistance shall be tested in accordance with ASTM D2794 using a 1.59-millimeter (1/16-inch) diameter punch and a 0.91-kilogram (2-pound) drop weight. The drop height shall be not less than 14 inches or the tester height corresponding to 3.16 Newton-meters (28 inch-pounds). Three impacts shall be made per panel. Impact shall be made to the coated side of the panel. The results shall conform to 3.6.4.

4.14.5 Slip resistance. Three test panels shall be prepared in accordance with 4.13.3 or in accordance with the requirements of the method approved by the qualifying activity (see 6.9). Slip resistance shall be tested in accordance with ASTM D2047 or a method approved by the qualifying activity (see 6.9) to determine conformance to 3.6.5.

4.14.6 Chemical resistance. Three test panels shall be prepared in accordance with 4.13.3. Chemical resistance shall be tested in accordance with ASTM D1308, spot test, covered procedure, at 23±2.5 °C (73±5 °F) for 24 hours, with the reagents listed below. The test shall begin no later than 12 hours from the cure-to-service time of the final coat as specified by the ASTM F718 product data sheet. Examinations shall be made at the start time, 2 hours, and 12 hours following chemical removal and shall conform to 3.6.6.

a. Reagent water (in accordance with ASTM D1193, type IV)

b. 10W-40 oil (in accordance with SAE J300)

c. JP-5 jet fuel (in accordance with MIL-DTL-5624)

4.14.7 AFFF resistance (type III only). Two test panels shall be prepared in accordance with 4.13.3 and tested in accordance with 4.14.6 for 30 days with the additional reagents listed below. The panels shall be visually examined immediately following exposure to reagents and 24 hours after air-drying to determine conformance to 3.6.7. The hardness shall also be evaluated 24 hours after air-drying, in accordance with ASTM D3363 to determine conformance to 3.6.7.

a. AFFF (in accordance with MIL-PRF-24385, type 3)

b. AFFF (in accordance with MIL-PRF-24385, type 6)

4.14.8 Salt spray (fog) resistance (type II only). Three test panels shall be prepared in accordance with 4.13.3. The panel shall be exposed to 5 percent salt spray for 14 days, in accordance with ASTM B117. Upon removal, the panels shall be washed gently in warm running water (not more than 38 °C [100 °F]) until free from any visible salt deposits and then examined. The results shall conform to 3.6.8.

4.15 Test procedures for types IV and V.

4.15.1 Chemical-resistant monolithic floorings. Test samples shall be prepared in accordance with ASTM C722. If test panel preparation is not specified in ASTM C722, panels shall be prepared in accordance with 4.13.5 for type V deck coverings or 4.13.6 for type IV deck coverings. The deck coverings shall be tested in accordance with ASTM C722 (except slip resistance). Slip resistance shall be tested in accordance with ASTM D2047 or a method approved by the qualifying activity (see 6.9). The results shall conform to the requirements of 3.7.1 (chemical-resistant monolithic floor surfacing) and 3.7.2 (slip resistance).

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4.15.2 Chemical resistance. Test samples shall be prepared in accordance with ASTM C722. When test panels are required, they shall be prepared in accordance with 4.13.5 for type V deck coverings or 4.13.6 for type IV deck coverings. The deck covering shall be tested in accordance with ASTM C722, and the results shall conform to the requirements of 3.7.1. The deck covering shall be tested in accordance with ASTM D1308, spot test, covered procedure, for 24 hours at 23±2.5 °C (73±5 °F) with the reagents listed below. The test shall begin no later than 12 hours from the covering cure-to-service time specified by the ASTM F718 product data sheet. Examinations, including 60-degree specular gloss in accordance with ASTM D523, shall be made at the start time, 2 hours, and 12 hours following chemical removal and shall conform to 3.7.3.

a. 10W-40 oil (in accordance with SAE J300)

b. JP-5 jet fuel (in accordance with MIL-DTL-5624)

c. AFFF (in accordance with MIL-PRF-24385, type 3)

4.15.3 Corrosion resistance. One test panel shall be prepared in accordance with 4.13.5 for type IV and type V deck coverings and tested in accordance with ASTM D1308, immersion procedure, for 15 days at 23±2.5 °C (73±5 °F) in substitute ocean water (ASTM D1141). Examination shall occur immediately following removal from reagent, and the results shall conform to the requirements of 3.7.4.

4.15.4 Impact resistance. Three test panels shall be prepared in accordance with 4.13.7 for type V deck coverings or 4.13.8 for type IV deck coverings. Impact resistance shall be tested by dropping a steel ball vertically from a height of 2.4 meters (8 feet) so that the impact shall be approximately at the center of the specimen. The steel ball shall weigh no less than 0.91 kilogram (2 pounds) and no more than 1.3 kilograms (3 pounds) and shall have a diameter no greater than 8 centimeters (3 inches). Impact shall be made to the covered side of the panel. The impact results shall conform to the requirements of 3.7.5.

4.15.5 Indentation resistance. Three test panels shall be prepared in accordance with 4.13.5 for type V deck coverings or 4.13.6 for type IV deck coverings. Deck covering systems shall be permitted to cure for 96 hours or the cure-to-service time specified by the ASTM F718 product data sheet, whichever is shorter. The thickness of the deck covering shall be measured in five locations not less than 2.54 centimeters (1 inch) from any edge of the specimen and not less than 6.4 centimeters (2.5 inches) from the center of an adjoining measurement. The thickness of the deck covering at each of these five locations shall be measured with a micrometer dial gage with a 113-gram (4-ounce) weight and a 6.4-millimeter (¼-inch) diameter flat foot, to an accuracy of ±0.76 millimeter (±0.03 inch). A permanent marker shall be used to mark these five locations. Each panel shall be mounted on a plate designed to support both the panel and a 907-kilogram (2,000-pound) continuous load without allowing the specimen to slide. The specimen mounting plates selected for the indentation tests shall be checked for flatness before being used. Indentations shall be made on the deck covering material at each of these five locations by placing a load of 907 kilograms (2,000 pounds) over a flat-faced, circular indenter, for 30 minutes at each spot. The indenter's flat face shall have an area of 6.5 square centimeters (1 square inch). Care shall be taken to ensure that the specimen is maintained parallel and level to the ground throughout the 30-minute indentation period. The results shall conform to the requirements of 3.7.6.

4.15.6 Vertical sag (type V only). The bottom 10.2 centimeters (4 inches) of one test panel shall be prepared in accordance with 4.13.5 for type V deck coverings or 4.13.6 for type IV deck coverings. Immediately following application of each part of the deck covering system, the panel shall be oriented in a vertical position (90 degrees) on a horizontal support surface with the bare area of the panel above the application area for 24 hours at 25±2 °C (73±5 °F). The thickness results of a cross-section of the specimen shall conform to the requirements of 3.7.7.

4.15.7 Abrasion resistance. Three test panels shall be prepared in accordance with 4.13.9 for type V deck coverings or 4.13.10 for type IV deck coverings. The deck covering shall be tested in accordance with ASTM D4060 using a CS-17 wheel and a 1-kilogram (2.2-pound) load for 1,000 cycles. The results shall conform to the requirements of 3.7.8.

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4.16 Test procedures for type VI.

4.16.1 Indentation resistance. Three test panels shall be prepared in accordance with 4.13.11. The deck covering indentation resistance shall be tested to determine conformance to 3.8.1. The test panels shall be prepared at the minimum thickness identified according to the ASTM F718 product data sheet. The location of three indentations shall be identified on each sample. The center of each indentation shall be not less than 2.54 centimeters (1 inch) from the edge of the specimen and not less than 2.54 centimeters (1 inch) from the center of the adjoining indentation. A load of 907 kilograms (2,000 pounds) shall be applied to the deck covering for 30 minutes by means of a circular indenter with a cross-sectional area of 6.5 square centimeters (1 square inch) and a radius of 0.5 millimeter (1/64 inch) on the perimeter of the indenting flat face. The thickness of the deck surfacing shall be measured by means of a thickness gage before and after a 907-kilogram (2,000-pound) load has been applied; the thickness of the steel base plate shall be subtracted in each case. The indentation shall be taken as the percent change in the thickness of the material. The results shall conform to the requirements of 3.8.1.

4.16.2 Impact resistance. Three test panels shall be prepared in accordance with 4.13.11. The deck covering shall be tested to determine conformance to 3.8.2. Each test panel shall be tested separately while being firmly supported on a solid horizontal base. A steel ball weighing no less than 3.5 kilograms (7.75 pounds) and no more than 4 kilograms (8.75 pounds) shall be dropped vertically from a height of 2.4 meters (8 feet) so that the impact shall be approximately at the center of the specimen. Each specimen shall be subjected to two impacts of the steel ball. The results shall conform to the requirements of 3.8.2.

4.16.3 Moisture absorption. The deck covering shall be tested in accordance with ASTM C413, method B to determine conformance to 3.8.3. The weight gain shall be reported as a percentage of moisture gain in accordance with ASTM C413. The results shall conform to the requirements of 3.8.3.

4.16.4 Bond strength. Six panels shall be prepared in accordance with 4.13.12. Specimens shall be tested by measuring the load required to shear the 5.1-centimeter by 5.1-centimeter (2-inch by 2-inch) area of deck covering from the steel plate by means of a shear test jig, as shown on figure 1. Three panels shall be tested to determine the initial bond strength in shear, and three panels shall be tested after aging at a temperature of 70 °C (158 °F) for 96 hours. The average bond strength shall be the average of the six shear loads expressed in kilograms per square centimeter (pounds per square inch) of applied deck covering system. The results shall conform to the requirements of 3.8.4.

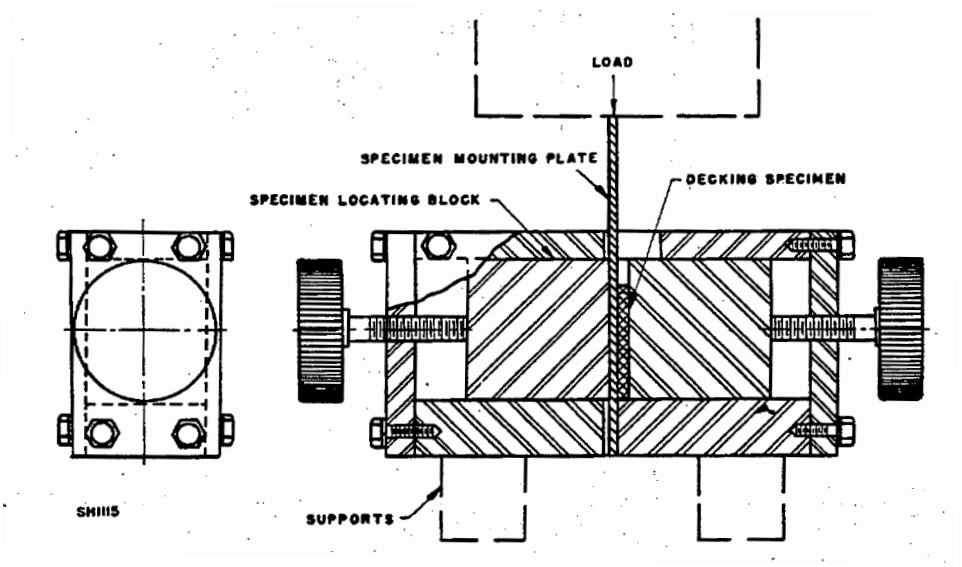


FIGURE 1. Shear test jig for hard-setting deck covering materials.

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4.16.5 Live load resistance. Three test specimens shall be prepared in accordance with 4.13.13. Each specimen shall be mounted on the live load apparatus (see 4.16.5.1) and subjected to the following loading procedure, and the results shall conform to the requirements of 3.8.5:

a. A roller load of 181 kilograms (400 pounds) shall be applied and initial thickness readings taken at four equally spaced positions of the roller as it passes over the specimen.

b. A roller load of 454 kilograms (1,000 pounds) shall be applied and the specimens subjected to 25 cycles of reciprocation under this load.

c. The roller load shall be increased in 226-kilogram (500-pound) increments after 25 cycles applied at each loading, until a maximum roller load of 1,361 kilograms (3,000 pounds) is attained.

d. At the end of the loading procedure, the final thickness of each specimen shall be measured at the four equally-spaced positions with the roller load reset at 181 kilograms (400 pounds). The average decrease of thickness for each specimen is then determined by the difference between the initial and final thickness readings.

4.16.5.1 Live load apparatus. The live load apparatus shall consist of a reciprocating table that supports the test specimen of the material under test and a roller that is brought to bear against the specimen by a pneumatic cylinder having a piston diameter of 20 centimeters (8 inches). The axis of the roller shall be parallel to the plane of the specimen and perpendicular to the specimen's line of motion, so that the roller rotates as the specimen reciprocates under it. The test specimen table shall oscillate at approximately 9 cycles per minute through a stroke of 12.7 centimeters (5 inches). The roller shall be brass, 10 centimeters (4 inches) in diameter and 8 centimeters (3⅛ inches) wide. The roller contacting force shall be controlled by varying the pressure in the loading cylinder.

4.16.6 Slip resistance. One test panel shall be prepared in accordance with 4.13.11 or in accordance with the requirements of the method approved by the qualifying activity (see 6.9). Slip resistance shall be tested in accordance with ANSI A326.3, ASTM D2047, or a method approved by the qualifying activity (see 6.9) to determine conformance to 3.8.6.

4.16.7 Sparking resistance. One test panel shall be prepared in accordance with 4.13.11. The deck covering shall be tested by abrading the specimen with a rotating, stiff steel wire wheel attached to an electric or pneumatic drill and by striking the specimens with glancing blows by a steel hammer. These tests shall be run in a darkened room, a dimly-lit room, a red-light-lit room, or other conditions where the presence or absence of sparks shall be noted. The results shall conform to the requirements of 3.8.7.

4.16.8 Electrical resistance. Three test panels shall be prepared in accordance with 4.13.11. The deck covering shall be tested in accordance with ASTM D257 to determine conformance to 3.8.8, utilizing the following:

a. Electrode: The electrode shall be 6.35 centimeters (2.5 inches) in diameter, weigh 2.27 kilograms (5 pounds), and have a rubber pad base of 55 durometer hardness cemented to one face. The contact surface of the electrode shall be covered with a sheet of aluminum foil held to the sides of the electrode with elastic bands.

b. Method of measurement: Direct-reading instrument.

c. Applied voltage: 500 volts.

4.16.9 Abrasion resistance. Three test panels shall be prepared in accordance with 4.13.14 so that an area of approximately 40 square centimeters (6 square inches) is presented to the abrasive in the machine specified in 4.16.9.1. The abrasive grit shall be No. 80 aluminum oxide. The specimen shall then be subjected to 1,500 revolutions of the wear test machine and the thickness obtained at the same points previously measured. The difference in thickness shall be averaged to determine the thickness of material worn away by the abrasive. The results shall conform to the requirements of 3.8.9.

4.16.9.1 Wear test machine. The wear test machine shall meet the following essential mechanical characteristics:

a. Mild steel, abrasion disk, 36 centimeters (14.2 inches) in diameter and 13 millimeters (½ inch) thick, revolved clockwise at a constant speed of 23.5 revolutions per minute.

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b. Specimen holder, 50 millimeters by 101 millimeters (2 inches by 4 inches), attached to a 16-millimeter (⅝-inch) diameter shaft by means of a universal coupling. The distance between centers of the disk and the specimen holder shaft is 130 millimeters (5.1 inches). A specimen of decking material is secured in the specimen holder by means of clamps at each end of the holder. The holder revolves clockwise at a constant speed of 32.5 revolutions per minute.

c. Hardened tool-steel cam attached to the specimen holder shaft by means of set screws. In each half revolution of the shaft, the specimen is lifted a distance of 1.6 millimeters (1/16 inch) and then dropped back on the disk specified in 4.16.9.1.a under a 4.5-kilogram (10-pound) weight. Change in the lift due to wear of the specimen is compensated by adjustment of the cam at regular intervals, so that the lift is maintained at a practically constant distance.

d. Weight of 4.5 kilograms (10 pounds), the maximum weight for this test machine.

e. Distributor for the abrasive grit. The grit flow by gravity from the hopper specified in 4.16.9.1.f through a 5-millimeter (⅕-inch) inside diameter tube and into the center of the distribution chamber. An impeller in the chamber provides an evenly distributed feed of the grit through eight 1.25-millimeter (1/20-inch) diameter holes centered on a 41-millimeter (1.6-inch) diameter pitch circle. A uniform distribution of grit falls to the disk specified in 4.16.9.1.a in the path of the test specimen. The distributor is revolved at a constant speed of 9 revolutions per minute.

f. Hopper for feeding abrasive grits

g. Bevel drive gear

h. Predetermining revolution counter and switch

i. 246-watt (one-third horsepower) compound wound direct current motor

j. Hopper for collecting used abrasive grit

k. Steel frame

4.16.9.2 Procedure. The steel disk specified 4.16.9.1.a revolves while the weighted specimen of deck covering in the holder specified in 4.16.9.1.b is lifted and dropped by a cam while it is being revolved.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Materials covered by this document are intended for use as deck coverings on U.S. Navy ships. Type I coatings are abrasion resistant and are for use on interior, dry, walking surfaces. Type II coatings, also abrasion resistant, are corrosion resistant and are for use on interior, wet, walking surfaces. Type III coatings are AFFF resistant and are for use on the deck of AFFF stations. Type IV deck coverings are for use on interior walking surfaces with an underlayment. Type V deck coverings are for use on interior walking surfaces without an underlayment. Type VI concrete is for use in areas requiring spark resistance, such as magazines. Class 1 deck coverings are not required to be off-gas tested and may be used aboard surface ships. Class 2 deck coverings are required to be off-gas tested and may be used aboard both surface ships and submarines. Grade A deck coverings must have a VOC content not greater than 250 grams per liter (2.08 pounds per gallon). Grade B deck coverings must have a VOC content not greater than 340 grams per liter (2.84 pounds per gallon). The resin chemistry is used to determine the composition.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, grade, and composition (see 1.2).

c. Requirement for ASTM methods, other consensus standards, or a copy of unique test methodology or any other necessary information on methodology used to determine reported ID characteristics (see 3.4).

d. Manufacturer's directions for use (ASTM F718 product data sheet) (see 3.5).

e. When conformance inspection is required (see 4.3).

f. Additional fire testing provisions (see 4.6.1 and 6.5).

g. Metals content (see 4.9).

h. When formulation data may be used by manufacturers in lieu of testing to demonstrate HAP content compliance (see 4.11).

i. Packaging and labeling requirements (see 5.1 and 6.10).

j. When a Safety Data Sheet (SDS) is required (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32584 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 SDS. When specified (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with Appendix D of 29 CFR 1910.1200. In order to obtain the SDS, federal acquisition regulation (FAR) clause 52.223-3 will be in the contract. The contracting activity should be given an SDS at the time of contract award.

6.5 Additional fire testing provisions. NAVSEA reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.6 Material certification. NAVSEA applies controls to materials like decking systems to be installed in submarines to mitigate the risk that volatile compounds released from the materials will adversely affect the atmosphere in a submarine. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510 AB ATM 010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by NAVSEA, or Type Commanders, or shipbuilders, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request must include descriptions of the material, the proposed application method for the material, and comments regarding whether or not the material will be stored on the submarine. The chemical constituents of the material are reviewed to determine if off-gas testing is required. Off-gas testing is conducted in a Government approved laboratory. NAVSEA reviews the chemical constituents of the material and the off-gas test results to determine the usage category for the decking system.

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6.7 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

6.8 Clarification of terms. Within this specification, the coating "components" refers to the individual coating components (e.g., resin, hardener) that are mixed for application and cure of the coating. The system "parts" refers to the individual coats or broadcast material (e.g., primer, topcoat, base/color coat, chips/aggregate, sealer coat) that make up the coating or deck covering system.

6.9 Alternations to the slip resistance method. The qualifying activity may allow slip resistance testing to be performed in accordance with ASTM C1028 or other acceptable methods if requested. Other test panel sizes may be used as required for the utilized test method as approved by the qualifying activity.

6.10 Packaging recommendations. Suggested packaging requirements for direct Government acquisitions are contained in table III (see 6.2).

TABLE III. Packaging.

Packaging	Recommended requirements for direct Government acquisitions
Containers	The components should be furnished in cans or bags appropriate to kit requirements capable of holding 0.47 liter (1 pint), 0.945 liter (1 quart), 3.78 liters (1 gallon), and 18.9 liters (5 gallons). Kits should consist of multiple containers, one of which should be large enough to contain all components for mixing purposes. Not all components are required to be supplied in a single can.
	Multiple friction plug containers should be in accordance with PPP-C-96, type V, class 2. Interior coatings should be as specified therein. Wire handles, as specified therein, should be provided for the 3.78-liter (1-gallon), or greater, containers. Closure of the properly filled and sealed cans should be as specified in the appendix thereto.
	All shipping containers should comply with the requirements of the Uniform Freight Classifications (UFC), the National Motor Freight Classification (NMFC), and the applicable requirements of 49 CFR and all United Nations (UN) Recommendations on the Transport of Dangerous Goods - Model Regulations.
	Unit of procurement: the deck covering materials covered by this specification should be purchased by volume. The unit of procurement should be in multiples of 1 liter or 1 U.S. liquid gallon at 15.5 °C (60 °F).
Packaging to ASTM D3951	Packaging considered commercial in nature should conform to the requirements of ASTM D3951.
	All containers should comply with the requirements of the UFC, the NMFC, and the applicable requirements of 49 CFR and all UN Recommendations on the Transport of Dangerous Goods - Model Regulations.
Level of pack (Packaging should be specified as follows)	Overseas delivery packaging. Intermediate containers of like-size kits of deck covering material should be packed in close-fitting wood boxes conforming to ASTM D6251/D6251M, overseas type, or ASTM D6880/D6880M, class 2. Box closure and strapping should be as specified in the applicable box specification or the appendix thereto, except that strapping should be flat and the finish B.
	Domestic delivery (level B) packing. Level B packing should be the same as for level A, except that boxes should be domestic type or class and the strapping should be finish A or B.

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TABLE III. Packaging – Continued.

Packaging	Recommended requirements for direct Government acquisitions
Palletization	Intermediate containers should be palletized in accordance with MIL-STD-147.
Packing for acquisitions involving direct delivery to Navy ships or installations	Treated lumber and plywood: All lumber and plywood, including laminated veneer materials, used in shipping container and pallet construction, member, blocking, bracing, and reinforcing must be fire-retardant treated material in accordance with MIL-L-19140 as follows: a. General use, weather resistant: MIL-L-19140, type II, category 1. b. General use, non-weather resistant: MIL-L-19140, type I, category 1.
Unit kits	The deck covering materials covered by this specification should be packed and packaged as kits.
SDS and product/procedure data sheets	A copy of the SDS and company product data/procedure sheets should be attached to the shipping document for each destination (see 6.4).
Hazardous warnings	Labels should be in accordance with 29 CFR Parts 1910, 1915, 1917, 1918, 1926, and 1928, and 49 CFR.
	All individual containers should have the following marking (except type VI): "CAUTION: This material contains volatile solvents, with probable hazardous vapors. Use with adequate ventilation. Avoid prolonged breathing of vapors or spray mists. The solvents are highly flammable; avoid open flame and smoking."
	Each component container, shipping container, and palletized load should be marked with the appropriate hazardous symbol.
	Unit containers should be marked: "This product is asbestos, lead, chromium, and cadmium free, except for possible trace levels." Type VI should also be marked: "This product is free of volatile organic hazardous air pollutants (VOHAPS/HAPS), except for possible trace levels."
VOC	VOC content should be stated on the label using the following statement: "Contains (insert VOC content) grams per liter (insert VOC content in grams per liter [pounds per gallon]) of volatile organic content per 40 CFR 60, Appendix A-7, Method 24."
HAP	HAPs should be stated on the label using the following statement: "Contains (insert HAP content) grams per liter (insert HAP content in grams per liter [pounds per gallon]) solids (nonvolatiles) per 40 CFR 63."
Package stability of paint (shelf life)	Labels should include the product manufacturer date, shelf life expiration date, and should meet the requirements of MIL-STD-129.
Application instructions	Manufacturers of decking materials are required to include the ASTM F718 product data sheet with each shipment of the material covered by this specification.

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6.11 Subject term (key word) listing.

Cement
Coating
Color flake
Concrete
Cosmetic polymeric
Deck paint
Fire performance
High durability
Light-weight concrete
Mastic
On-deck insulation
PRC
Terrazzo

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH

Preparing activity:
Navy − SH
(Project 8010-2021-006)

Review activities:
Army – MI, MR, TE
Navy – AS, CG, MC
DLA – CD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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